

Home Lab Project Week 4, Day 4

Task

This task is about familiarizing with Wireshark, capturing interfaces, display filters, and coloring rules, and practicing capturing network traffic.

Task: Familiarize yourself with Wireshark capture interfaces, display filters, and coloring rules, and practice capturing network traffic.

1. Identify the capture interface

Start Wireshark in Kali:

wireshark

Look at the available interfaces and select the interface connected to your isolated lab network.

Your Kali VM should use:

Kali: 192.168.64.2

Ubuntu: 192.168.64.3

Network: 192.168.64.0/24

To confirm Kali's interface from the terminal, I type:

ip addr

Select the corresponding interface in Wireshark and start a capture.

2. Generate some basic traffic

From Kali, generate traffic toward Ubuntu:

```
ping 192.168.64.3
```

Let it run for several seconds, then press:

Ctrl + C

You should see **ICMP Echo Request** and **ICMP Echo Reply** packets in Wireshark.

This gives you your first simple traffic capture for the home lab.

3. Practice display filters

In Wireshark's **Display Filter** field, try:

```
icmp
```

This displays only ICMP traffic.

You can also try:

```
ip.addr == 192.168.64.3
```

This filters traffic involving your Ubuntu system.

Another useful filter is:

```
ip.src == 192.168.64.2
```

This shows traffic originating from Kali.

Important: Display filters do not delete packets. They simply control which captured packets are displayed.

4. Practice coloring rules

Wireshark uses colors to make different types of traffic easier to recognize.

Go to:

View → Coloring Rules

Review the existing rules and observe how different packets are visually distinguished.

For Day 4, you don't need to create complicated custom rules. The objective is simply to understand how coloring helps identify traffic quickly.

5. Save your capture

After generating traffic, stop the capture and save it as a **.pcapng** file.

For example:

week4_day4_basic_capture.pcapng

You could keep your captures organized in the directory created on Day 3:

~/cybersecurity-lab/captures/

This capture can become evidence for your project documentation.

Evidence to collect

For your Day 4 documentation, I recommend taking screenshots of:

1. **Wireshark capture interface** showing the active lab interface.
2. **Captured ICMP traffic** generated between Kali and Ubuntu.
3. **icmp display filter** showing filtered packets.
4. **ip.addr == 192.168.64.3 filter** showing Ubuntu-related traffic.
5. **Coloring Rules window**.
6. **Saved .pcapng capture file**.

How Week 4 now fits together

Day	Activity	Purpose
Day 1	Network mapping + VM snapshots	Verify connectivity and establish recovery baseline
Day 2	Linux command-line practice	Prepare Kali for cybersecurity activities
Day 3	Install/configure Wireshark	Prepare the packet-capture environment
Day 4	Wireshark interfaces, filters, coloring + capture	Practice actual packet capture

So **Day 4 completes the Wireshark setup/practice phase**. Your next traffic-analysis tasks can then build on the **.pcapng** capture you created here.